

Exact Analytic Orbits of a Nonlinear p -Adic Double Shear: Minimal Components and Every Residue Period

HCS-C394 theorem and reproducibility package

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Abstract

A nonlinear symplectic double shear on the p -adic unit polydisc admits an analytic extension of its original integer iteration clock. We give the joint Tate-algebra construction at every nonzero parameter with valuation at least one for odd primes and at least two at the prime two. The origin is its only genuine periodic point. A strict first-term estimate gives the exact two-time displacement scale p^{-c-2r} on the radius- p^{-r} shell. Every nonzero orbit closure is therefore an isometric copy of the p -adic adding-one system. The same identity determines all least periods, shell multiplicities and fixed counts at every prime-power residue level. Independent exact computations audit the formulas without replacing their proofs. The interpolation and analytic-zero methods have classical owners; no new certificate algorithm or target prime-period identity is claimed.

Keywords: p -adic dynamics; symplectic shear; analytic interpolation; minimal component; residue period; algebraic return.

中文摘要

本文研究单位多圆盘上的非线性辛双剪切映射，将原始整数迭代时间联合解析地延拓到素进整数。奇素数下参数估值至少为一，二进情形至少为二，参数零单独排除。原点是完整素进空间内唯一真正的周期点。首项严格支配给出精确的双时间位移尺度，从而识别每个非零最小轨道闭包，并统一确定所有素数幂剩余层的最小周期、循环重数与固定点数。精确有限计算只检验公式的实现。经典插值理论明确归属，不将原生算术结构写成目标零点对应。

关键词：素进动力学；辛剪切；解析插值；最小分量；剩余周期；代数返回。

1 One nonlinear source and its arithmetic boundary

Let p be any rational prime and normalize $v_p(p) = 1$. Fix a nonzero $a \in \mathbb{Z}_p$ with $c = v_p(a) \geq 1$ when p is odd and $c \geq 2$ when $p = 2$. On $X = \mathbb{Z}_p^2$ use the maximum norm and freeze

$$F_a(x, y) = (x + ay^2, y + a(x + ay^2)^2). \quad (1)$$

The coefficientwise proximity to the identity permits analytic time interpolation, but proximity alone does not describe all orbits. For this map, the additional homogeneous shear structure determines the exact radius-dependent clock. Finite residue permutations and periodic points in X must remain separate throughout.

Poonen's joint interpolation theorem and its dyadic boundary [1] supply the classical method reconstructed below. The analytic zero-set approach to dynamical Mordell–Lang belongs to Bell–Ghioca–Tucker [2]. Recent finite-precision and refined-tail certificate methods belong to Mukhamedov [3]; we claim no new effective certification algorithm. Our question is the full orbit structure of the particular nonlinear source, not a new general interpolation theorem.

The nearest project mechanisms are C166's finite affine dyadic Pascal tower, C174's expanding parity renewal, C258's mixed congruential map, and C283's conductor-shell heat operator. None supplies the nonlinear joint analytic action and valuation-scale decomposition of (1). The symbolic branch's P23 imports classical linear-recurrence zero-set rigidity, not this nonlinear analytic-arc proof. These comparisons define the repository increment; they are not a global novelty certification.

2 Symplecticity, reversal, and joint analytic time

Write $F_a = S_y S_x$ for the two shears $S_x(x, y) = (x + ay^2, y)$ and $S_y(x, y) = (x, y + ax^2)$. Both preserve $dx \wedge dy$, and

$$F_a^{-1}(x, y) = (x - a(y - ax^2)^2, y - ax^2). \quad (2)$$

The involution $R_0(x, y) = (-y, -x)$ satisfies $R_0 S_x R_0 = S_y^{-1}$ and $R_0 S_y R_0 = S_x^{-1}$, hence $R_0 F_a R_0 = F_a^{-1}$ on the same clock. Integral coefficients in (1)–(2) make both maps 1-Lipschitz; each is therefore an isometry and induces a permutation modulo every p^N .

Theorem 1. *There is $G_a \in \mathbb{Z}_p\langle x, y, t \rangle^2$ such that $G_a(u, n) = F_a^n(u)$ for every integer $n \geq 0$ and*

$$G_a(G_a(u, s), t) = G_a(u, s + t) \quad (s, t \in \mathbb{Z}_p).$$

Thus negative time is the inverse of positive time, without a new roof.

Proof. On $A = \mathbb{Z}_p\langle x, y \rangle$, set $\Delta h = h \circ F_a - h$. Factoring differences of monomials and taking Gauss-norm limits gives $\|\Delta h\|_p \leq p^{-c} \|h\|_p$. In particular $\Delta^m u \in p^{cm} A^2$. The joint series

$$G_a(u, t) = \sum_{m \geq 0} \binom{t}{m} \Delta^m u \quad (3)$$

converges in the Tate algebra because its m th term has coefficient norm at most $p^{-cm + v_p(m!)}$ and $v_p(m!) \leq m/(p-1)$. If $\delta = c - 1/(p-1) > 0$, its tail after M has norm at most $p^{-(M+1)\delta}$. At integer $n \geq 0$ the series truncates to $(I + \Delta)^n u = F_a^n(u)$. The action law holds first at nonnegative integer pairs. Such pairs are dense in $\mathbb{Z}_p^2$ and all compositions are continuous and integral-valued, so the law holds at every pair. The time-zero map is the identity. $\square$

The sufficient threshold cannot be replaced by a general dyadic coefficient congruence modulo two: $f(x) = -x$ has a nonzero two-cycle. An analytic interpolant of the orbit of one would equal one on all even times and therefore identically, contradicting its odd-time value. This general control does not assert failure of every double shear outside the stated threshold. Pointwise residue identity also does not suffice: $x^p = x$ on $\mathbb{F}_p$, but $x^p - x$ has unit coefficients.

3 The exact displacement scale and minimal decomposition

Lemma 1. *If $u \neq 0$ and $r = \min(v_p(x), v_p(y))$, then for every $s, t \in \mathbb{Z}_p$,*

$$\|G_a(u, t) - G_a(u, s)\|_p = p^{-c-2r}|t-s|_p. \quad (4)$$

Proof. First take a unit-radius vector v and parameter b of valuation d . The vector $F_b(v) - v = b(v_2^2, (v_1 + bv_2^2)^2)$ has norm p^{-d} : either v_2 is a unit or $v_1 + bv_2^2$ is a unit. For $m \geq 2$,

$$\binom{t}{m} = \frac{t}{m} \binom{t-1}{m-1}, \quad v_p(m) < d(m-1).$$

The binomial value on the right is integral, by continuity from integer values. The strict inequality holds for odd $p, d \geq 1$; the only possible equality at $d = 1$ is $p = m = 2$, excluded by $d \geq 2$ in the dyadic case. The m th higher term in (3) therefore has value norm at most $|t|_p p^{-d-1}$, whereas the first term has norm exactly $|t|_p p^{-d}$. Strict dominance gives $\|G_b(v, t) - v\|_p = p^{-d}|t|_p$. This is a value estimate, not the coefficient Gauss estimate used in Theorem 1.

Now write $u = p^r v$. Substitution gives $F_a(p^r v) = p^r F_{ap^r}(v)$; interpolation and continuity give the same identity for every t . Since $d = c + r$, $\|G_a(u, t) - u\|_p = p^{-c-2r}|t|_p < \|u\|_p$. Thus $G_a(u, s)$ has the same radius as u . Apply the last formula at that basepoint with time $t - s$ and use the action law. This proves (4), with both sides zero when $s = t$. $\square$

Theorem 2. *Every nonzero orbit closure is the image $G_a(u, \mathbb{Z}_p)$, a scaled isometric copy of $\mathbb{Z}_p$ on which F_a is conjugate to addition by one. These closures are pairwise disjoint or equal and, with the origin, partition X into minimal compact invariant sets. The origin is the only genuine periodic point and $DF_a^n(0) = I$ for every $n \geq 1$.*

Proof. Equation (4) gives injectivity and an inverse continuous on the compact image. Density of nonnegative integers identifies this image with the forward orbit closure. The action law conjugates iteration to addition by one, which meets every residue class and is minimal. A common point and negative time identify any two intersecting images. No nonzero integer n can make the right side of (4) zero. The derivative at the origin follows directly from (1). $\square$

4 Every finite residue period and its native zeta

For $N \geq 1$ and $0 \leq r < N$, define

$$L_{N,r} = p^{\max(0, N-c-2r)}, \quad S_{N,r} = p^{2(N-r)} - p^{2(N-r-1)}.$$

Theorem 3. *A nonzero residue vector modulo p^N with minimum valuation r has exact period $L_{N,r}$. Shell r has $S_{N,r}/L_{N,r}$ cycles. The zero residue is fixed. For every integer $n \geq 1$,*

$$\# \text{Fix}(F_a^n \bmod p^N) = p^{2(N-R)}, \quad R = \max\left(0, \left\lceil \frac{N-c-v_p(n)}{2} \right\rceil\right). \quad (5)$$

Proof. The minimum valuation $r < N$ is independent of the lift. The displacement formula says that return modulo p^N is equivalent to $c + 2r + v_p(n) \geq N$, proving the least period. Subtract the number of vectors divisible by p^{r+1} from those divisible by p^r to count the shell. All its cycles have the same length, so division gives their count. The shells with $r \geq R$ and the zero class telescope to (5). In particular, every shell with $c + 2r \geq N$ is fixed, not only the zero class. The zero class and all shells together have exactly p^{2N} points. $\square$

The complete finite point-counting product is

$$\zeta_N(z) = (1-z)^{-1} \prod_{r=0}^{N-1} (1 - z^{L_{N,r}})^{-S_{N,r}/L_{N,r}}. \quad (6)$$

Its logarithmic derivative recovers (5). When distinct shells share a period their cycle counts add. By contrast, the genuine zeta on X is only $(1-z)^{-1}$. Finite cycles have p -power lengths, but their nonzero representatives never become genuine finite-period points.

Revision record

Round one: strict first-term dominance, all minimal components, every residue period and the complete finite native zeta are added with proofs.

References

- [1] B. Poonen, *p-adic interpolation of iterates*, Bull. Lond. Math. Soc. **46** (2014), 525–527. doi:10.1112/blms/bdu010.
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